

ON THE HOMOGRAPHIC TRANSFORMATION OF A SURFACE OF THE SECOND ORDER INTO ITSELF.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 326–333.]

The following theorems in plane geometry, relating to polygons of any number (odd or even) of sides, are, well known.

“If there be a polygon of  $(\alpha + 1)$  sides inscribed in a conic, and so if the sides pass through given points, the  $(\alpha + 1)$ th side will envelope a conic having double contact with the given conic.” And “If there be a polygon of  $(\alpha + 1)$  sides inscribed in a conic, and so of the sides touch conics having double contact with the given conic, the  $(\alpha + 1)$ th side will envelope a conic having double contact with the given conic.” The second theorem of course includes the first, but it is the two separately that I propose to extend to a surface of the second order.

As regards the corresponding theory in geometry of three dimensions, Sir W. Hamilton has given a theorem relating to polygons of an odd number of sides, which may be thus stated: “If there be a polygon of  $(2\alpha + 1)$  sides inscribed in a surface of the second order, and  $2\alpha$  of the sides pass through given points, the  $(2\alpha + 1)$ th side will constantly touch two surfaces of the second order, each of them intersecting the given surface of the second order in the same four lines<sup>1</sup>.”

The entire theory depends upon what may be termed the transformation of a surface of the second order into itself, or analytically, upon the transformation of a quadratic form of four indeterminates into itself. I use for shortness the term transformation simply; but this is to be understood as meaning a homographic transformation, or in analytic language, a transformation by means of linear substitutions. It will be convenient to remark at the outset, that if two points of a surface of the second order have the relation contemplated in the data of Sir W. Hamilton’s theorem (viz. if the line joining the two points pass through a fixed point), the transformation is, using the language of the *Recherches Arithmétiques*, an *improper* one, but that the relation contemplated in the conclusion of the theorem (viz. that of two points of a surface of the second order, connected by a line touching two surfaces of the second order each of them intersecting the given surface of the second order in the same four lines) depends upon a *proper* transformation; and that the circumstance that an even number of improper transformations is required in order to make a proper transformation (that this circumstance, I say), is the reason why the theorem applies to polygons in which an even number of sides pass through fixed points, that is, to polygons of an *odd* number of sides.

Consider, in the first place, two points of a surface of the second order such that the line joining them passes through a given point. Let  $x, y, z, w$  be current coordinates<sup>2</sup>, and let the

<sup>1</sup>See *Phil. Mag.* vol. xxxix. (1851), p. 396. The form in which the theorem is exhibited by Sir W. Hamilton is somewhat different; the surface considered as being not ellipsoid, and the two surfaces touched by the last or  $(2\alpha + 1)$ th side of the polygon are spoken of as having focal connection with two confocal hyperboloids of two sheets, having respectively double contact with the given ellipsoid; the student is, in fact, a quadric conic which is taken as an analytical premise. (The conclusion is necessarily of the kind mentioned; but when this is once understood, the language of the text may be properly employed. It may be proper to explain that hyperboloids of one sheet inscribed in an ellipsoid is, by analogy with the case in plano of the hyperbola of imaginary transformation, the ulterior expression of the doubled conic of a surface of the second order than is implied.

<sup>2</sup>Strictly speaking, it is the *ratios* of these quantities,  $x, y, z, w$ , which are the coordinates, and consequently, even when the point is given, the values  $x, y, z, w$  are merely indeterminate to a factor *près*. So that in assuming that a point is given, we should write  $x : y : z : w = a : b : c : d$  and so on, but when this is once understood, the language of the text may be properly employed. It may be proper to explain that the equation  $(a, b, c, d \mid x, y, z, w)^2 = Rxx + yy^2$ , where “ $\mid$ ” means  $a(x + b(y + c)z + dw)$ . The system of coefficients may frequently be indicated by a single coefficient only; thus in the text  $(a, \dots)(x, y, z, w)^2$  stands for the most general quadratic function of four variables.

equation of the surface be

$$(a, \dots \rangle x, y, z, w)^2 = 0,$$

and take for the coordinates of the two points on the surface  $x_1, y_1, z_1, w_1$  and  $x_2, y_2, z_2, w_2$ , and for the coordinates of the fixed point  $\alpha, \beta, \gamma, \delta$ . Write for shortness

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta)^2 = p,$$

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta \rangle x, y, z, w) = q_1,$$

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta \rangle x_2, y_2, z_2, w_2) = q_2,$$

then the coordinates  $x_2, y_2, z_2, w_2$  are determined by the very simple formula

$$\begin{aligned} x_2 &= x_1 - \frac{2q_1}{p} y_1, \\ y_2 &= y_1 - \frac{2q_1}{p} y_1, \\ z_2 &= z_1 - \frac{2q_1}{p} y_1, \\ w_2 &= w_1 - \frac{2q_1}{p} y_1. \end{aligned}$$

In fact, the point  $(\xi, \eta, \zeta, \omega)$  lies in the line joining the points  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$ ; and to show that this line touches the surface  $V = 0$ , it is only necessary to form the equation of the tangent plane at the point  $(\xi, \eta, \zeta, \omega)$  to the surface in question; this is

$$(\alpha, \dots \rangle (x, y, z, w \rangle \xi, \eta, \zeta, \omega)^2 = 0,$$

that is, the point  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$  will correspond to each other by means of the above formulæ, and that they are determined by means of the equations  $\xi, \eta, \zeta, \omega$  are given by the  $x_1 + x_2, y_1 + y_2, z_1 + z_2, w_1 + w_2$ .

Suppose now that we consider three points 1, 2, 3 of the surface, the lines joining 1, 2 and 2, 3 passing respectively through given points; we have first

$$\begin{aligned} x_2 &= x_1 - \frac{2q_1}{p} \alpha, & y_2 &= y_1 - \frac{2q_1}{p} \beta, \\ z_2 &= z_1 - \frac{2q_1}{p} \gamma, & w_2 &= w_1 - \frac{2q_1}{p} \delta, \end{aligned}$$

where  $q_1, \beta, \gamma, \delta$  are the parameters which may subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable conic may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)]\beta = 0,$$

Or, what is the same thing, the system of equations is

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0, \quad \xi + \eta = c\xi + c\eta$$

$$(2 - c)\alpha^2 - p(\alpha + c\eta)\alpha + c\eta\xi + (-c\eta + p + c\eta)\eta + (-c\eta - \mu c + c\eta)\eta = 0.$$

or, it may also be written,

$$(a^2 + p + a^2 c\eta + (a\eta - \mu)c + c\eta a^2)\alpha + (b\eta - \beta c)\eta + c(-c\eta + \mu_0)\eta + (-c\eta - p - c\eta + a\eta)\eta = 0.$$

Represent for a moment the form  $\eta\eta - \mu c + \mu c\eta$  by  $V$ , and consider the surface  $V = 0$ . If we may also be written,

$$(\alpha^2 + \beta^2 + c\eta + (\eta\eta - c\eta + b\eta)c)\eta + p\alpha\eta = 0,$$

this discriminant is

$$(\eta + V\eta + \alpha\eta + V\eta + \theta^2,$$

which shows that the surface intersect in four lines. Suppose the discriminant vanishes; we have for the determination of  $\eta$  a quadratic equation, which may be written

$$q'^2 + (2 + s)q' + R = 0;$$

but the roots of this equation by  $q'$ ,  $q''$ , then each of the functions  $\eta + Vc'$ ,  $\eta + Vc''$  will break up into linear factors, and we may write

$$\eta + Vc'_r = R_r R_r,$$

$$\eta + Vc''_r = R'_r R_r,$$

( $V'$  and  $V''$  are of course linear functions of  $R_r$ , and  $R_r$ ,  $R'_r$ .) These which lead in evidence the fact of the two surfaces intersecting in four lines.

The equations

$$q_r + q_r = Rq_r, \quad q_r = q_r = 2q_r, \quad \alpha_r + \alpha_r = Rq_r, \quad \&c.$$

generating lines and in like manner in the point  $Q$ . Thus  $P$ ,  $Q$  are tangent planes at the points  $A$ ,  $B$  respectively, and the system of these polygons of generating lines is the system of common chords. The chord in question is a tangent to the two surfaces involved; it is therefore the line of intersection of the tangent planes through  $P$  to the same  $PA$  as belonging determinately to one or the other of the two systems of generating lines, then when  $P$  is given, the corresponding point  $Q$  is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines of intersection of  $\theta$ ,  $\phi$ , and then the proper transformation is obtained is not the general proper transformation.

It is necessary to remark, that, whenever the coordinates of the points  $Q$  are connected with the coordinates of the points  $B$  by means of the equations which belong to an improper transformation, the points  $P$ ,  $Q$  have to each other the geometrical relation above mentioned, viz. there exist two plane sections  $\theta$ ,  $\phi$  such that  $P$ ,  $Q$  are the opposite angles of a skew quadrangle upon the surface, and having the other two angles upon  $\theta$  and  $\phi$ .  $Q$  is the opposite angles of a skew quadrangle upon the surface, having two of its opposite angles upon  $\theta$  and  $\phi$  respectively, and the two adjacent angles upon  $\theta$  and  $\phi$ .

It is, perhaps, hardly possible to embrace by a single intelligible synthesis the remaining side will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

I conclude with the remark, that we may by means of two plane sections of a surface of the second order obtain a proper transformation. For, if the generating lines through  $P$  meet the sections  $\theta$ ,  $\phi$  in the points  $A$ ,  $B$  respectively, and the remaining generating lines through  $A$ ,  $B$  respectively meet in the points  $\phi$ ,  $\theta$  respectively in  $B'$ ,  $A'$ , and the remaining generating lines through  $B'$ ,  $A'$  respectively meet in a point  $P'$ ; then will  $P$ ,  $P'$  be a pair of corresponding points in a proper trans-formation.

*2 Stone Buildings, January 11, 1854.*

## ON THE GEOMETRICAL REPRESENTATION OF AN ABELIAN INTEGRAL.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 414–418.]

The equation of a surface passing through the curve of intersection of the surfaces

$$ax^2 + by^2 + cz^2 + dw^2 = 0,$$

$$ax^2 + b'y^2 + c'z^2 + d'w^2 = 0,$$

is of the form

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

where  $k$  is an arbitrary parameter. Suppose that the surface touches a given plane, we have for the determination of  $k$  a certain equation the root of which may be considered as parameters defining the plane in question. Let one of the values of  $k$  be considered equal to a given quantity  $k$ , the plane touches the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and the other two values of  $k$  may be considered as parameters defining the particular tangent plane, or what is the same thing, determining the point of contact with the surface.

Or more clearly, then—in order to determine the position of a point on the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

the tangent plane at the point in question is touched by two other surfaces

$$p(ax^2 + by^2 + cz^2 + dw^2) + q(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

$$p'(ax^2 + by^2 + cz^2 + dw^2) + q'(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and similarly for the other lines.

I consider now a curve of double curvature ( $k$ ) touching the surface ( $k$ ); for the points of the curve ( $k$ ), the parameter of one of the surfaces touched at the point in question is constant, that is,  $p, q$  are functions of the parameters  $p', q'$  of the other; or what is the same thing, writing  $k = p \div q$ ,  $k' = p' \div q'$ , the coordinates of the point ( $k$ ) of the curve ( $k$ ) are

$$x : y : z : w = \sqrt{(\alpha + k)(\alpha + k')} : \sqrt{(\beta + k)(\beta + k')} : \sqrt{(\gamma + k)(\gamma + k')} : \sqrt{(\delta + k)(\delta + k')},$$

or under a somewhat more convenient form we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

where  $a, b, c, d$  are inversely as  $a', b', c', d'$ . Taking for a moment  $k = k'$ , this is a thing thus, viz. it is

$$x : y : z : w = (a + k) : (b + k) : (c + k) : (d + k),$$

which equation is satisfied by writing  $(a + k)x : (b + k)y : (c + k)z : (d + k)w$ , where  $\alpha, \beta, \gamma, \delta$  are inversely as  $a, b, c, d$ . Similarly, we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

for the conditions in order that the point  $(k, k')$  may lie in the tangent plane at  $(k')$ ,  $p, q$  to the surface  $(k')$ . Similarly, we have

$$\sum (b+k)\sqrt{(c+k)(c+k')} + (c+k)\sqrt{(b+k)(b+k')} = 0,$$

for the condition in order that the point  $(k, p)$  may lie in the tangent plane at  $(k', p, q)$  to the surface  $(k)$ . Similarly,

$$\sum (a+k')(a+k)\sqrt{(a+k)(a+k') + (b+k')(b+k)\sqrt{(b+k)(b+k')}} = 0,$$

or making the same reductions give

$$\sum (a+a')\sqrt{(a+k)(a+k')} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b'',$$

and the like for the system of equations

$$\sqrt{(a+a')(a+\rho)(a+\sigma)(a+\tau)} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b''.$$

When in each system  $a$  is to be successively replaced by  $a, b, c, d$ , and  $\rho, \sigma, \tau$  and  $\mu, \nu$  are indeterminate. Now dividing each equation of the one system by the corresponding equation in the other system, we see that the equation

$$\frac{a+b}{a+a'} h + \frac{1}{\mu+a+b''} = h+a,$$

is satisfied by the values  $a, b, c, d$  of  $a$ , and therefore, since the equation is in  $a$  only of the third order, that the equation in question must be identically true. We may therefore write

$$h + \mu + a + a'' = (a+a)(\lambda + \mu),$$

and the two systems of equations become identical with

$$\sqrt{(a+k)(a+k')(a+p)(a+q)(a+r)(a+s)} = h(a+a)(\rho + \mu).$$

A set of remarks which may be represented by the single equation

$$\Phi(x+y)\Phi(x+y) = \mu(x+y) + (\nu x + \nu y)(x+y),$$

where  $x$  is arbitrary; or in the same thing, writing  $\theta = a+x, a+b, b+b$ ,

$$x(x+a)(x+b)(x+c)(x+d)(x+\alpha)(x+\beta) = h(\mu+a)(\rho+\mu).$$

Hence, putting the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

$$\Pi_1 p = \int \frac{p dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}},$$

we see that the algebraical equations between  $p, q, p', q'$  are equivalent to the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

The algebraical equations which connect  $\theta, \phi$  with  $p, q, p', q'$ , may be exhibited under several different forms; thus, for instance, considering the point  $(k, k')$  as a point in the line joining  $(k, p)$ , and  $(k', q')$ , the equation is

$$\begin{vmatrix} \sqrt{(a+p)(a+q)} & \sqrt{(a+k)(a+k')} & \sqrt{(a+p')(a+q')} \\ \sqrt{(b+p)(b+q)} & \sqrt{(b+k)(b+k')} & \sqrt{(b+p')(b+q')} \\ \sqrt{(c+p)(c+q)} & \sqrt{(c+k)(c+k')} & \sqrt{(c+p')(c+q')} \\ \sqrt{(d+p)(d+q)} & \sqrt{(d+k)(d+k')} & \sqrt{(d+p')(d+q')} \end{vmatrix} = 0,$$

i.e. the determinants formed by selecting any three of the four columns must vanish; the equations so obtained are equivalent (as they should be) to two independent equations.

Or, again, by considering  $(p, q, \phi)$  first as a point in the tangent plane at  $(k, k')$  to the surface  $(k)$ , and then as a point in the tangent plane at  $(k', p', q')$  to the surface  $(k')$ , we obtain

$$\begin{aligned} \sum (a+k) \sqrt{(a+p)(a+q)} \sqrt{(a+k)(a+k')} &= 0, \\ \sum (a+k') \sqrt{(a+p)(a+q)} \sqrt{(a+k)(a+k')} &= 0. \end{aligned}$$

Or again, we may consider the line joining  $(k, k')$  and  $(p, q)$ , the formulæ will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

The condition in order that the line joining the points  $(k, k', p, q)$  and  $(k', k'', p', q')$  may meet a curve which is represented by the single equation

$$\sum_a a x_a x_b x_c \cdots = 0,$$

may also be expressed in the form

$$2k\zeta + (\Pi p + \Pi q + \Pi p' + \Pi q') = \text{const.},$$

the equation entails

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

or

$$\sum \frac{dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}} = \text{const.}$$

considering only the first integral, the second one being analogously.

The preceding formulæ are obviously very incomplete; but the connection which they point out between the geometrical question and the Abelian integral involving the root of a function of the sixth order, may I think be of service in the theory of these integrals.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 427–431.]

M. St Laurent has shown (*Gergonne*, vol. xviii. [1827] p. 1), that in certain cases the caustic by refraction of a circle is identical with the caustic of refraction of a circle (the refracting circle and radiant point being, of course, properly chosen), and a very elegant demonstration of M. St Laurent's theorems is given by M. Gergonne in the same volume, p. 48. A similar method may be employed to demonstrate the more general theorem, that the same caustic by refraction of a

circle may be considered as arising from six different systems of a radiant point, circle, and index of refraction. The demonstration is obtained by means of the secondary caustic, which (as is well known) an oval of Descartes. Such oval has three rectangular axes, and two corresponding circles and indices of refraction, as follows:—

Let  $x$  be the radius of the refracting circle,  $\xi, \eta$  be the coordinates of the radiant point, and the secondary caustic in the envelope of the circle

$$x^2 + \beta^2 = c^2,$$

where  $\alpha, \beta$  are parameters which vary subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable circle may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)]\beta = 0,$$

which is of the form

$$U + \lambda + B(\lambda^2 + \delta) = 0,$$

the envelope is therefore

$$U^2 = A^2(\lambda + B),$$

Hence substituting, we have for the equation of the envelope, i.e. for the secondary caustic,

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 = 4c^2[(xx - \xi)^2 + (yy - \eta)^2],$$

and this may perhaps be considered as the standard form.

To show that this equation belongs to a Descartes' oval, suppose for greater convenience  $\eta = 0$ , and write

$$\rho^2(x^2 + y^2) = \rho^2 - \xi^2 + 2c\eta\sqrt{kx - \xi\xi} + a^2,$$

multiplying this equation by  $1 - \frac{1}{\alpha}$ , and adding to each side  $a^2\left(x - \frac{\xi}{a}\right) + (x - \xi)^2 + y^2$ , we have

$$\begin{aligned} & \left(1 - \frac{1}{\alpha}\right) [\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 + \left(x - \frac{\xi}{a}\right)^2 \\ &= (x - \xi)^2 + y^2 + 2c\left(x - \frac{1}{a}\right) \sqrt{(x - \xi)^2 + y^2} + c^2 \left(x - \frac{1}{a}\right)^2, \end{aligned}$$

or reducing

$$\alpha^2 \left[ \left(x - \frac{\xi}{a}\right)^2 + y^2 \right] = \left[ \sqrt{(x - \xi)^2 + y^2} + c \left(x - \frac{1}{a}\right) \right]^2;$$

again, multiplying the same equation by  $\frac{1}{a^2} \left(1 - \frac{1}{a^2}\right)$ , and adding to each side  $\frac{1}{a^2} \left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2} y^2 + \xi^2 + y^2$ , we have

$$\frac{1}{a^2} \left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2} y^2 + (x - \xi)^2 + y^2 + 2\frac{c}{a} \left(1 - \frac{\xi}{a}\right) \sqrt{(x - \xi)^2 + y^2} + \frac{c^2}{a^2} \left(1 - \frac{\xi}{a}\right)^2,$$

or reducing,

$$\left(x - \frac{\xi}{a}\right)^2 + y^2 = \frac{1}{a^2} \left[ \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(1 - \frac{\xi}{a}\right) \right]^2.$$

Hence, extracting the square roots of each side of the equations thus found, we have the equation of the secondary caustic in either of the forms

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} = \frac{1}{a} \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(x - \frac{1}{a}\right),$$

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} = \frac{1}{a} \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(1 - \frac{\xi}{a}\right),$$

to which are to be joined

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} \sqrt{\left(x - \frac{\xi}{a'}\right)^2 + y^2} + \frac{c}{a} \left(x - \frac{1}{a'}\right) + \frac{c}{a} \left(1 - \frac{\xi}{a}\right) \sqrt{\left(x - \frac{\xi}{a'}\right)^2 + y^2} + \frac{c}{a'} \left(1 - \frac{\xi}{a}\right) \sqrt{1 - a\xi^2 + y^2} =$$

Write successively,

$$\xi' = \xi, \quad c' = c, \quad \mu' = \mu; \quad (1)$$

$$\xi' = \frac{\xi}{a}, \quad c' = \frac{c}{a}, \quad \mu' = \frac{1}{\mu}; \quad (\alpha)$$

$$\xi' = \frac{1}{\xi}, \quad c' = \frac{c}{\xi}, \quad \mu' = \frac{\mu}{\xi}; \quad (\beta)$$

$$\xi' = \xi, \quad c' = c, \quad \mu' = \frac{1}{\mu}; \quad (\gamma)$$

$$\xi' = \frac{\xi}{a}, \quad c' = \frac{c}{a}, \quad \mu' = \mu; \quad (\delta)$$

$$\xi' = \frac{1}{\xi}, \quad c' = \frac{c}{\xi}, \quad \mu' = \frac{1}{\mu}; \quad (\epsilon)$$

or, what is the same thing,

$$\xi' = \xi, \quad c' = c, \quad \mu' = \mu; \quad (1)$$

$$\xi' = \xi, \quad c' = \xi, \quad \mu' = \xi/\mu; \quad (\alpha)$$

$$\xi' = \xi, \quad c' = \xi/\mu, \quad \mu' = \mu; \quad (\beta)$$

$$\xi' = \xi, \quad c' = c, \quad \mu' = 1/\mu; \quad (\gamma)$$

$$\xi' = c/\mu, \quad c' = c, \quad \mu' = \mu/c; \quad (\delta)$$

$$\xi' = c/\mu, \quad c' = c, \quad \mu' = 1/\mu; \quad (\epsilon)$$

then, whichever system of values of  $\xi$ ,  $c$ ,  $\mu$  be substituted for  $\xi$ ,  $c$ ,  $\mu$ , we have in each case identically the same secondary caustic, the effect of the substitutions being simply to interchange the different forms of the equation; and we have therefore identically the same caustic. By writing

$$(\xi, c, \mu) \rightarrow (\xi', c', \mu') = *(\xi, c, \mu),$$

where  $\alpha, \beta, \gamma, \delta, \epsilon$  will be functional symbols, such as are treated of in my paper "On the Theory of Groups as depending on the symbolic equation  $\theta^n = 1$ ," [125], and it is easy to verify the equations

$$1 = \alpha\beta = \beta\alpha = \gamma^2 = \delta\epsilon,$$

$$\alpha = \beta\gamma = \gamma\beta = \delta^2 = \epsilon\delta,$$

$$\beta = \alpha\gamma = \gamma\alpha = \epsilon\delta = \delta\epsilon,$$

$$\gamma = \alpha\beta = \beta\alpha = \delta\epsilon = \epsilon\delta,$$

$$\delta = \alpha\beta = \beta\alpha = \gamma\delta = \delta\gamma,$$

$$\epsilon = \alpha\delta = \delta\alpha = \gamma\epsilon = \epsilon\gamma.$$



Suppose, for example,  $\xi = c$ , i.e. let the radiant point be in the circumference; then in the fourth system  $\xi' = c$ ,  $c' = c$ ,  $\mu' = \frac{1}{\mu}$ , i.e. since the radius of the refracting circle is the radius of the circle and the index of refraction reciprocal, this is one of M. St Laurent's theorems, viz.

THEOREM. The caustic by refraction of a circle when the radiant point is in the circumference, is the same caustic by refraction for the same radiant circle and a reciprocal index of refraction. The first of these is a group of eighteen, viz.

Again, let  $\xi = c\mu$ , the fifth system gives  $\xi' = \frac{c}{\mu}$ ,  $c' = c$ ,  $\mu' = -1$ , or the new system is in this case also a reflecting system. This is the other of M. St Laurent's theorems, viz.

THEOREM. The caustic by reflection of a circle when the distance of the radiant point from the centre is equal to the radius of the circle multiplied by the index of refraction, is the caustic by reflection of the same circle for a radiant point which is the image of the first radiant point.

Of course it is to be understood that the image of a point means a point whose distance from the centre is equal to square of radius  $\div$  distance.

*2 Stone Buildings, Nov. 2, 1853.*

125.

ON THE THEORY OF GROUPS, AS DEPENDING ON THE SYMBOLIC EQUATION  $\theta^n = 1$ .

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 40–47.]

Let  $\theta$  be a symbol of operation, which may, if we please, have for its operand, not a single quantity  $x$ , but a system  $(x, y, \dots)$ , so that

$$\theta(x, y, \dots) = (x', y', \dots),$$

where  $x', y', \dots$  are any functions whatever of  $x, y, \dots$ , it is in not even necessary that  $x', y', \dots$  should be the same in number with  $x, y, \dots$ . In particular  $x', y', \dots$  may represent a permutation of  $x, y, \dots$ .  $\theta$  is in this case what is termed a substitution; and if, instead of a set  $x, y, \dots$ , we consider points or magnitudes (or, even, operations) any how related, then  $\theta$  may be an operator giving rise to a substitution, or it may be in particular a substitution, in like manner. The general question of the analytical representation of a substitution does not enter into the present plan, but the symbols  $\theta, \phi$ , etc.,  $\theta, \theta'$ , etc. are interpreted in such manner that the symbols  $\theta\phi, \phi\theta$ , etc. have a definite signification, but this need not always be the case; for instance, in the theory of matrices, the symbols  $\theta, \phi$ , etc. do not in general satisfy the commutative law  $\theta\phi = \phi\theta$ , so that,  $\theta\phi, \phi\theta$ , etc. are to be considered as in general different symbols. In other respects also the analogy is by no means perfect, and the words "operation" and "operand" are used as the only ones really suitable; the theory of operators, or as it may be called, the theory of operations, in the theory of analytical operations functional process which gives rise to the symbols in question, may be considered analogous to or having the same meaning as multiplied together so as to form a table, thus:—

| Further Symbols |          |               |               |     |
|-----------------|----------|---------------|---------------|-----|
|                 | 1        | $\alpha$      | $\beta$       | ... |
| 1               | 1        | $\alpha$      | $\beta$       | ... |
| $\alpha$        | $\alpha$ | $\alpha^2$    | $\alpha\beta$ | ... |
| $\beta$         | $\beta$  | $\beta\alpha$ | $\beta^2$     | ... |
| $\vdots$        | $\vdots$ | $\vdots$      | $\vdots$      |     |

that as well rows as columns of the square will contain all the symbols  $1, \alpha, \beta, \dots$ . It follows that when there is no entry of any symbol of the group without repetition, and by such order

whatever, is a symbol of the group. Suppose that this group

$$1, \alpha, \beta, \gamma, \dots,$$

contains a symbol  $\beta$  which may be such that there exists in the same group  $\theta^n = 1$ .<sup>3</sup>

satisfies the equation  $\theta^n = 1$ , where  $n$  is the smallest number for which this is the case, then we may speak of  $\theta$  as a composite symbol of any order  $n$  less than  $n$  a composite number. It follows then by a like reasoning as for  $\beta, \gamma$ , the system  $1, \alpha, \beta, \gamma$  is a subgroup of  $\theta$ . It follows from this, in particular, that there can be no system in which any of the others, but are associative,  $\theta(\phi\chi) = (\theta\phi)\chi$ . The expression of the symbol of an operation  $\theta$  is given by the symbol  $\theta = a + b\alpha + c\beta + \dots$ , the form of  $\theta = a + b\alpha + c\beta + \dots$ , the symbols at the same time being known are commutative or  $\theta\phi = \phi\theta$ , the system is said to be Abelian. There may be involved THEOREM. — If the symbols of the group be multiplied together two and two in any order, the entire system being known are not necessarily distinct. It may be done by means of the substitution equation  $\theta^n = 1$ , where  $n$  is some positive integer; this is the case of a system of all the roots of the equation  $x^n = 1$ , in the same notation, the symbols of the equation are

$$1, \alpha, \alpha^2, \dots, \alpha^{n-1} \quad (\alpha^n = 1).$$

This definition explains the meaning of the equation  $\theta^n = 1$ ; viz. this means therefore represent the group symbols, viz.

$$1, \alpha, \beta, \gamma, \dots, \quad (\beta^n = 1),$$

thus multiplying each term by a further factor, we have for the group  $1, \alpha, \alpha\beta, \alpha\alpha\beta$ , but we must be equal either to 1 or to one of the roots. In the former case our group is —

$$1, \alpha, \beta \quad (\alpha^n = 1).$$

which is analogous to the system of roots of the ordinary equation  $\theta^n - 1 = 0$ . For the sake of comparison with what follows, I remark, that representing the last- mentioned group by

$$1, \alpha, \beta, \gamma,$$

we have the table

|          | 1        | $\alpha$ | $\beta$  | $\gamma$ |
|----------|----------|----------|----------|----------|
| 1        | 1        | $\alpha$ | $\beta$  | $\gamma$ |
| $\alpha$ | $\alpha$ | $\beta$  | $\gamma$ | 1        |
| $\beta$  | $\beta$  | $\gamma$ | 1        | $\alpha$ |
| $\gamma$ | $\gamma$ | 1        | $\alpha$ | $\beta$  |

If, on the other hand,  $\alpha^2 = 1$ , then it is easy by similar reasoning to show that we must have  $\beta = \beta\alpha$ , so that the group is of the form

$$1, \alpha, \beta, \alpha\beta, \quad (\alpha^2 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha),$$

or if expressed under the group by

$$1, \alpha, \beta, \gamma,$$

we have the table

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<sup>3</sup>The idea of a group as applied to permutations or substitutions is due to Galois, and the introduction of it may be considered as marking an epoch in the progress of the theory of algebraical equations.

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|          | 1        | $\alpha$ | $\beta$  | $\gamma$ |
|----------|----------|----------|----------|----------|
| 1        | 1        | $\alpha$ | $\beta$  | $\gamma$ |
| $\alpha$ | $\alpha$ | 1        | $\gamma$ | $\beta$  |
| $\beta$  | $\beta$  | $\gamma$ | 1        | $\alpha$ |
| $\gamma$ | $\gamma$ | $\beta$  | $\alpha$ | 1        |

or, if we please, the symbols are such that

$$\alpha^2 = \beta^2 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta, \quad \beta = \alpha\gamma = \gamma\alpha, \quad \gamma = \alpha\beta = \beta\alpha,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation  $x^n - 1 = 0$ .]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if  $u$  be the parameter, and

$$\alpha(u) = \frac{u}{u}, \quad \beta(u) = (-u), \quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have  $\alpha = \beta$ , and therefore  $\gamma = 1$ , in which point of view this group is simply a group of two symbols 1,  $\alpha$  ( $\alpha^2 = 1$ ).

Again, in the theory of matrices, if  $I$  denote the operation of inversion, and  $\text{tr}$  that of transposition, ( $I$  do not stop to explain in terms as the example may be passed over), we may write

$$\alpha = I, \quad \beta = \text{tr}, \quad \gamma = I \cdot \text{tr} = \text{tr} \cdot I.$$

I proceed to the case of a group of six symbols,

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

which may be considered as representing a system of roots of the symbolic equation

$$\theta^6 = 1.$$

It is in the first place to be shown that there is at least one root which is a prime root of  $\theta^6 = 1$ , or (to use a simpler expression) a root having the index 3. It is clear that if there were a simpler expression a root having the index 2. It is true that if the index 3,  $\theta$  must be done by means of this root would have the index 2; then in the square of  $\theta$  would also have this index, that is, it would have the index 6 which is the cube root of the symbol. The symbol  $\theta = 1$  may always be considered as having the index 6, considered in the symbol  $\theta = 1$ , the index 3 of a root having the index 3. It is impossible that *all* the roots should have the index 3. This may be done by means of the simpler observation: if so, then, since their product has also the index 2 in the symbol  $\theta = 1$ , this is to have the index 6.

Assume that the symbols  $\alpha, \beta, \gamma$  have the index 3. The square of any one of this root, e.g.  $\alpha$ , is then  $\alpha^2$ , and these are contained in the group, must have either the index 6 or the index 2 (or to be a simpler expression) a root having the index 3, having each of the order 4. It is easy to see that any of the roots of  $\theta$ , and that the group of the order 4. It is easy to see that any of the roots of  $\theta$ , and that the group is, in fact,  $1, \gamma\delta, \delta\gamma, \delta\gamma^2$ , which is, in fact, the group of six having the index 3, having obtained ; I prefer commencing with the assumption of a root having the index 3. Suppose that  $\alpha$  is such a root, the group must clearly be of the form

$$1, \alpha, \alpha^2, \beta, \alpha\beta, \alpha^2\beta \quad (\alpha^3 = 1);$$

and multiplying the entire group by  $\gamma$  as a nearer factor, it becomes  $\gamma, \alpha\gamma, \alpha^2\gamma, \gamma', \alpha\gamma', \alpha^2\gamma'$ ; we must therefore have  $\beta\gamma = \alpha\beta$  or  $\gamma\beta = \alpha\beta$ . But the latter supposition  $\gamma' = \alpha\beta$  leads also to the same name in which case we may assume  $\gamma' = 1$ ; then we have  $\beta^2$  also equal to 1,  $\alpha$  or  $\alpha^2$ , but the first two cases lead to the supposition  $\gamma^2 = \alpha$  or  $\beta\gamma = 1$ , and to a system of six symbols  $1, \alpha, \alpha^2, \gamma\beta, \gamma\alpha\beta, \gamma\alpha^2\beta$ , when we must have either  $\gamma\alpha = \alpha\gamma$  or else  $\gamma\alpha = \alpha^2\gamma$ . The former assumption leads to the group

$$1, \alpha, \alpha^2, \gamma, \alpha\gamma, \alpha^2\gamma \quad (\alpha^3 = 1, \gamma^2 = 1, \alpha\gamma = \gamma\alpha),$$

which is, in fact, analogous to the system of roots of the ordinary equation  $x^n - 1 = 0$ . For the sake of comparison with what follows, I remark, that representing the last-mentioned group by the general symbols

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

we have the table

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
|------------|------------|------------|------------|------------|------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
| $\alpha$   | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ | 1          |
| $\beta$    | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    |
| $\delta$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    | $\gamma$   |
| $\epsilon$ | $\epsilon$ | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   |

while if we represent the second of these two forms, viz. the group

$$1, \alpha, \alpha^2, \gamma\beta, \gamma\alpha\beta, \gamma\alpha^2\beta, \quad (\alpha^3 = 1, \gamma^2 = 1, \gamma\alpha = \alpha^2\gamma),$$

by the same general symbols

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

we have the table

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
|------------|------------|------------|------------|------------|------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
| $\alpha$   | $\alpha$   | $\beta$    | 1          | $\delta$   | $\epsilon$ | $\gamma$   |
| $\beta$    | $\beta$    | 1          | $\alpha$   | $\epsilon$ | $\gamma$   | $\delta$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    |
| $\delta$   | $\delta$   | $\epsilon$ | $\gamma$   | $\alpha$   | $\beta$    | 1          |
| $\epsilon$ | $\epsilon$ | $\gamma$   | $\delta$   | $\beta$    | 1          | $\alpha$   |

or, what is the same thing, the symbols are such that

$$\alpha^3 = \beta^3 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta; \quad \beta = \alpha\gamma = \gamma\alpha,$$

$$\gamma = \delta\beta = \alpha\beta,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation  $x^n - 1 = 0$ .]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if  $u$  be the parameter,  $\alpha$  a fixed root, and tr denotes the operation of transposition, we have the group symbols which may be expressed in the form

$$\alpha(u) = \frac{u-c}{u-d}, \quad \beta(u) = (-u), \quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have  $\alpha\beta = \gamma$ , and although the small group  $\{1, \alpha, \beta, \gamma\}$  is simply a group of two symbols 1,  $\alpha$ , ( $\alpha^2 = 1$ ).

If we represent the second of two, and only two, essentially distinct forms of a group of six.

If we represent the second of these two forms, viz. the group

$$1, \alpha, \beta, \alpha^2, \beta\alpha, \beta\alpha^2,$$

$$\alpha = \beta, \gamma = \beta\alpha, \delta = \beta\alpha^2,$$

and we have thus two, and only two, essentially distinct forms of a group of six.

126.

ON THE THEORY OF GROUPS, AS DEPENDING ON THE SYMBOLIC EQUATION  $\theta^n = 1$ .  
SECOND PART.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 408–409.]

IMAGINE the symbols

$$L, M, N, \dots$$

such that ( $L$  being any symbol of the system),

$$L^{-1}L = LL^{-1} = N, LN = N, \dots$$

is the group

$$1, \alpha, \beta, \dots,$$

then, in the first place,  $M$  being any other symbol of the system,  $M^{-1}LM, M^{-1}M, \dots$  will be the same group  $1, \alpha, \beta, \dots$ . In fact, the system  $L, M, N, \dots$  may be written  $L, L\alpha, L\beta, \dots$ , and so the system  $M, \beta M, \gamma M, \dots$  will be  $M^{-1}LM, M^{-1}\alpha LM = (M^{-1}LM)\alpha M, \dots$  which belongs to the group  $1, \alpha, \beta, \dots$ .

Next it may be shown that

$$LL', ML', NL', \dots$$

is a group, although  $LL'$  may not in general be the same group as  $1, \alpha, \beta, \dots$ . In fact, writing  $M = L\alpha, N = L\beta, \dots$ , the symbols just written down are

$$LL', L\alpha L', L\beta L', \dots,$$

and we have e.g.  $L\alpha L' = L\alpha L' = L\sigma L', = L\sigma L'$ , where  $\sigma$  belongs to the group  $1, \alpha, \beta, \dots$ ;

[Pages 130: continuation in similar group-theory style omitted in full transcription.]

127.

ON THE HOMOGRAPHIC TRANSFORMATION OF A SURFACE OF THE SECOND ORDER INTO ITSELF.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 208–212; continuation of **122**.]

I pass to the original transformation. Sir W. R. Hamilton has given (in the note, p. 723 of his *Lectures on Quaternions* [Dublin, 1853]) the following theorem:—If there be a polygon of  $2\alpha$  sides inscribed in a surface of the second order, and  $(2\alpha - 1)$  of the sides pass through

given points, the side or  $(2\alpha - 1)$ th side will constantly touch two cones circumscribed about the surface of the second order. The relation between the extremities of the  $(2\alpha - 1)$ th side is that of two points connected by the general improper transformation; in other words, if there be a surface of the second order into itself, then have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

The tangent planes through  $P$  to the surface in question form an involution; the two sections which are the siliconugates of the involution pass through  $P$ , and having to consider the corresponding points  $\theta$ ,  $\phi$  in the same theorems, viz. that the two points which are the involution of  $\theta$  and  $\phi$ , the surface in the manner described in the result of  $PA$  as belonging determinately to one of the two systems of generating lines, then  $P$  is given, the corresponding point  $Q$  is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

I proceed to establish (by M. Hermite's method) the formulæ for the particular case in question. The thing required is to find  $x_2, y_2, z_2, w_2$  linear functions of  $x_1, y_1, z_1, w_1$ , such that

$$\begin{aligned} ax_2^2 + by_2^2 + cz_2^2 + dw_2^2 &= ax_1^2 + by_1^2 + cz_1^2 + dw_1^2, \\ (ay_1y_2 + bz_1z_2 + cz_1z_2 + dw_1w_2)^2 - (ax_1^2 + by_1^2 + cz_1^2 + dw_1^2)(ax_2^2 + by_2^2 + cz_2^2 + dw_2^2) \\ &= Lx_1x_2 + My_1y_2 + Nz_1z_2 + Sw_1w_2, \end{aligned}$$

where  $L, M, N, S$  are constants of course, in the case in question functions of the constants  $a, b, c, d$  of the surface, and the linear functions of  $x_1, y_1, z_1, w_1$  which are to be the values of  $x_2, y_2, z_2, w_2$ . Following the analysis, in which  $L = 4, M = 4$ , the following system of equations, in which  $(a, b, c, d)$  of the points minimums, and  $M = ab + cd = 2bc + da$ , are obtained.

Following M. Hermite's method, the formulæ are obtained.

128.

#### DEVELOPMENTS ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 339–345.]

I propose to develop some particular cases of the theorems given in my paper, "Correction of two Theorems relating to the Porism of the in-and-circumscribed Polygon" (*Phil. Mag.* vol. vi. [1853], 114). The two theorems are as follows:—

THEOREM. The condition that there may be inscribed in the conic  $U = 0$  an infinity of  $n$ -gons circumscribed about the conic  $V = 0$ , depends upon the developement in ascending powers of  $\xi$  of the square root of the discriminant of  $\xi U + V$ , viz. if this square root be

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + H\xi^7 + \dots,$$

then for  $n = 3, 5, 7$ , &c. respectively, the conditions are

$$|C| = 0, \quad \begin{vmatrix} C & D \\ D & E \end{vmatrix} = 0, \quad \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix} = 0, \text{ \&c. ;}$$

and for  $n = 4, 6, 8$ , &c. respectively, the conditions are

$$|D| = 0, \quad \begin{vmatrix} D & E \\ E & F \end{vmatrix} = 0, \quad \begin{vmatrix} D & E & F \\ E & F & G \\ F & G & H \end{vmatrix} = 0, \text{ \&c.}$$

THEOREM. In the case where the conics are replaced by the two circles which (taken in their respective conjugates) are equivalent to a polygon inscribed in the one and circumscribed about the other, the values of  $x$  which give the position of  $\xi$  on which the dimensions are replaced by the

$$\alpha^2 + \beta^2 - R^2 = 0, \quad (x - c)^2 + y^2 - r^2 = 0,$$

then the discriminant, the square root of which appears as before, is

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \dots$$

Write for a moment

$$A = R\xi + (\xi^2 + D\xi^3 + E\xi^4 + \dots = \sqrt{(1 + \alpha\xi)(1 + \beta\xi)(1 + \gamma\xi)}),$$

then the discriminant, the square root of which appears as before, is

$$(1 + \xi)r^2 = \alpha + \beta + \gamma - 3\alpha\beta,$$

then

$$\begin{aligned} A &= 1, \\ 2B &= a + b + c, \\ -3C &= a^2 + b^2 + c^2 - 3bc, \\ 16D &= a^3 + b^3 + c^3 - 3a(b^2 + c^2 + bc), \\ -128E^2 &= a^4 + b^4 + c^4 - 4(b^3 + c^3 + abc^2 + a^2bc) + 6a^2bc. \end{aligned}$$

To adapt these to the case of the two circles, we have to write

$$\alpha^2(1 + \alpha\xi)(1 + \beta\xi)(1 + \gamma\xi) + 4r^2 = \beta + r^2(1 + \beta)\xi^2 + \frac{1}{4}r^2\xi^4,$$

and therefore

$$\begin{aligned} \alpha^2 &= 1, \\ \alpha^2(\alpha + \beta + c) &= -\xi^2, \end{aligned}$$

values which satisfy identically  $\alpha^2 + y^2 + z^2 + w^2 = c^2 + (a^2 + \alpha) - \beta + 3\alpha\beta$ , belong to an improper transformation. We have thus obtained the proper transformation of the surface of the second order  $ax^2 + by^2 + cz^2 + dw^2 = 0$  into itself.

Returning for a moment to the equations which lead to the surface  $\Sigma y - \Sigma \omega = 0$ , it is easy to see that they without loss of generality, in the points equations  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$ , then,

The resulting system of equations is, in fact, identical with

$$ax_2^2 + by_2^2 + cz_2^2 + dw_2^2 = ax_1^2 + by_1^2 + cz_1^2 + dw_1^2,$$

and forming the values, the determined values of  $x_2, y_2, z_2, w_2$  are linear functions of  $x_1, y_1, z_1, w_1$  of the form

$$MM'z_2 = (aa' - bb' - cc' - dd')z_1 + (ab' + a'b + cd' - c'd)y_1 + (ac' + a'c - bd' + b'd)z_1 - (b'c - bc' + a'd + ad')w_1 + (bd' - b'd)$$

values which satisfy identically  $x_2^2 + y_2^2 + z_2^2 + w_2^2 = x_1^2 + y_1^2 + z_1^2 + w_1^2$ , and which belong, as is supposed, to an improper transformation. We have thus obtained the proper transformation of the surface of the second order  $ax^2 + by^2 + cz^2 + dw^2 = 0$  into itself.

*2 Stone Buildings, January 11, 1854.*

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED TRIANGLE,  
AND ON AN IRRATIONAL TRANSFORMATION OF TWO TERNARY QUADRATIC FORMS EACH  
INTO ITSELF.

[From the *Philosophical Magazine*, vol. vii. (1855), pp. 115–121.]

There is an irrational transformation of two ternary quadratic forms each into itself, based upon the solution of the following geometrical problem.

Given that the conic

$$lx + my + nz = 0$$

meets the conic

$$(a, b, c, f, g, h)(x, y, z)^2 = 0$$

in the points  $(x_0, y_0, z_0)$ ,  $(x_1, y_1, z_1)$ , to find the other point of intersection.

The solution is exceedingly simple. Take  $(a, b, c)$  for the coordinates of the other point of intersection, we must have identically with respect to  $x, y, z$ ,

$(a, \dots)(x, y, z)^2 - 2(\alpha, \dots)(x_0, y_0, z_0)(x, y, z) = (lx + my + nz)(\beta_0(x - x_0) + \beta_1(y - y_0) + \beta_2(z - z_0))$ ,  
to a constant factor *près*. Assume successively  $y = 0, z = 0; x = 0, z = 0; x = 0, y = 0$ , then we have

$$\begin{aligned} l : m : n &= y_0 z_0(\alpha x_0 + g z_0 + h y_0) - x_0(a x_0 + h y_0 + g z_0) : \\ & z_0 x_0(h x_0 + b y_0 + f z_0) - y_0(h x_0 + b y_0 + f z_0) : \\ & x_0 y_0(g x_0 + f y_0 + c z_0) - z_0(g x_0 + f y_0 + c z_0), \end{aligned}$$

or as they may be more simply written,

$$l : m : n = y_0 z_0(ax_0^2 - by_0^2 - cz_0^2 - 2fy_0z_0) : z_0x_0(\dots) : x_0y_0(\dots).$$

[Pages 146–147: development of the transformation theory.]

We have, then, identically

$lx_0(lx + my + nz) + xx_0(my - nz)(xz - yx_0) + y_0(mz - ny)(xy - zx_0) + lz_0(\dots) - hx_0z_0(\dots) = 0$ ,  
which shows that if  $\theta_2, \eta_2$  may be as well to give the corresponding solution of the problem.

[Pages 147–148: further reductions; the irrational transformation

$$\sqrt{lx + my + nz} \quad \text{and} \quad \sqrt{(a, \dots)(x, y, z)^2}$$

provide an irrational transformation of the two forms each into itself.]

I shall once consider  $l, m, n$  as given functions of  $x_0, y_0, z_0$ , satisfying identically the equations

$$lx_0 + my_0 + nz_0 = 0, \quad Nc + na = 0, \quad xx_0 + yy_0 + zz_0 = 0,$$

equations which express that the line  $lx + my + nz = 0$  is the tangent at the point  $(x_0, y_0, z_0)$  to the conic  $(a, b, c, \dots)(x, y, z)^2 = 0$ . And I shall take for  $\beta, \gamma$  the following constant values, viz. in the case of the point  $(x, y, z)$  we shall have for the values:

$$\theta_1 = \theta_2 - q_0, (2qa - bx), \quad \eta_1 = \eta_2 - q_0, (2q\beta - ay), \quad \zeta_1 = \zeta_2 - q_0, (2q\gamma - cz),$$

and similarly for the values  $\theta_2, \eta_2, \zeta_2$  in the same point. We obtain immediately

$$\theta_2^2 + \eta_2^2 + \zeta_2^2 = \theta_1^2 + \eta_1^2 + \zeta_1^2.$$

[Pages 149–150: continuation through the proof.]

The preceding investigations have been in my possession for about eighteen months.

2 Stone Buildings, April 18, 1855.



## DEUXIÈME MÉMOIRE SUR LES FONCTIONS DOUBLEMENT PÉRIODIQUES.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xix. (1854), pp. 193–208; Sequel to Mémoire t. x. (1845), **25**.]

Je vais essayer de développer ici les propriétés qui se rapportent aux transformations linéaires des polynômes des fonctions  $p, g, \Theta, Z$ , dont je me suis occupé dans le Mémoire sur les fonctions doublement périodiques que j'ai donné dans ce Recueil en 1845. Avant d'entrer en matière, je remarque que partant des expressions

$$\Omega = u + u', \quad \Upsilon = v + v',$$

des deux périodes, on a  $i = \sqrt{-1}$ , on obtient, en définissant

$$\Omega' = u\nu - u'\nu', \quad \Upsilon' = u'\nu' - u\nu',$$

les équations

$$\begin{aligned} \Omega \Upsilon' &= u\nu + u'\nu' = -(\nu\Omega - \nu'\Omega'), \\ \Upsilon \Omega' &= u\nu' + u'\nu = (\nu\Omega + \nu'\Omega'), \end{aligned}$$

et au moyen desquelles

$$B = \frac{\nu(\nu\Omega - \nu'\Omega')}{i\Upsilon \bmod. (\nu\nu - \nu'\nu')}, \quad B' = \frac{\nu'(\nu\Omega + \nu'\Omega')}{i\Upsilon \bmod. (\nu\nu' - \nu'\nu)},$$

des quantités  $B, B'$  on déduit les formules

$$B + \beta = \frac{\nu\Upsilon}{i \bmod. (\nu\nu' - \nu'\nu)}\Omega, \quad B - B' = \frac{\nu\nu'}{i \bmod. (\nu\nu' - \nu'\nu)}.$$

Je me mets attention aux transformations qui correspondent à des entiers impairs et premiers, et je suppose, de plus, que les transformations soient toujours propres, et régulières; c'est-à-dire qu'on doit avoir

$$\begin{aligned} (2k+1)\Omega_r &= \lambda\Omega + \mu\Upsilon, \quad \bmod. (\mu, \nu) < T, \\ (2k+1)\Upsilon_r &= \nu\Omega + \rho\Upsilon, \quad \bmod. (\nu, \rho) < T, \end{aligned}$$

où  $2k+1$  est un nombre impair et premier,  $\lambda, \mu, \nu, \rho$  sont des entiers impairs ou pairs, et cômptes de tel signe près,  $\lambda\rho - \mu\nu = 2k+1$ , et que les coefficients sont, savoir, soit signe près,

$$\lambda\rho - \mu\nu = 2k+1.$$

(condition pour que la transformation soit propre), et de même que les fonctions

$$\lambda \equiv 0, \quad \mu \equiv 0, \quad \bmod. 2,$$

$$\nu \equiv 0, \quad \rho \equiv 1.$$

(condition pour que la transformation soit régulière).

On trouve tout de suite

$$\Omega = \rho\Omega_r - \mu\Upsilon_r, \quad \Upsilon = -\nu\Omega_r + \lambda\Upsilon_r,$$

j'écris aussi

$$\Omega_r = u_r + u'_r, \quad \Upsilon_r = v_r + v'_r,$$

et je suppose que  $R_r, \beta_r$  soient des fonctions de  $u_r, u'_r, v_r, v'_r$  analogues aux fonctions  $B, \beta$ , de  $u, u', v, v'$ .

Cela étant, je tire d'abord de l'équation

$$(2k+1)(B+\beta)\Omega = \frac{\nu}{i \bmod. (\nu\nu - \nu'\nu')} \Omega^2,$$

au moyen de laquelle

$$(B+\beta)\Omega = \frac{\nu}{i \bmod. (\nu\nu' - \nu'\nu)} \Omega^2,$$

au transformant en

$$\begin{aligned} [(2k+1)(B_r+\beta_r)\Omega &= \frac{\nu_r}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} \Omega_r^2, \\ &= \frac{\nu_r}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} (\rho\Omega_r - \mu\Upsilon_r)^2, \\ &= \frac{\rho^2\nu - 2\rho\mu(\nu + \nu') + \mu^2\nu'}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} (\rho\Omega_r - \mu\Upsilon_r). \end{aligned}$$

[Pages 152–156: continuation; the second Mémoire develops the expression of  $J_rx, g_rx, Z_rx$  in terms of the original  $Jx, gx, Zx$ , the products

$$J_rx = e^{-\frac{1}{2k+1}(B_r-2k+1\cdot B)x\pi i} \prod \frac{J(x+s\psi_r)}{J(s\psi_r)},$$

the cases “Premier cas” ( $p_r = 1, q_r = 2q_\theta$ ) and “Deuxième cas” ( $p_r = 2k+1, q_r = 2$ ), and the corresponding  $\Omega_r, \Upsilon_r$  formulæ. The exposition continues in [157–208] of this same Mémoire 130.]

Cela fait voir que

$$e^{-i\pi x} = e^{-i\pi x},$$

le coefficient  $A$  étant donné au moyen de l'équation

$$A = \sum \frac{1}{(m, n)},$$

où la somme est prise, comme auparavant, entre les limites

$$\bmod. (m, n) < T, \quad T = \infty.$$

Mais il n'est pas permis d'écrire

$$A = \iint \frac{dx dy}{(m, n)^2}.$$

En effet, cette intégrale n'est pas le premier terme d'une suite dont il faudrait, pour obtenir un résultat exact, prendre deux termes; le second terme de la suite serait une intégrale prise le long de tout le contour, et le second terme ne serait pas, comme dans le premier cas, égale à zéro. Pour trouver la valeur de  $A$ , je remarque que les fonctions doublement périodiques  $p, g, \Theta, Z$ , qui sont contenues dans les formules ordinaires, peuvent satisfaire aux conditions

$$\bmod. (m, n) < T, \quad T = \infty,$$

puisqu'on a l'analyse  $T = x$ , et puis l'on obtient que la fonction  $Ly + My^2$ . Cela étant, on substituera pour  $u$  la valeur, je forme l'équation

$$\frac{y(x+s\psi)}{y(x)} = e^{-bx-cx^2} \Pi\left(1 + \frac{x}{(m, n)+s\psi}\right),$$

et j'écris successivement

$$y = \frac{1}{2}\Pi, \quad y = \frac{1}{2}T,$$

et qui donne pour les équations correspondante du produit infini double doublement  $e^{i\pi x}$ ,  $Gx$ ; en comparant les valeurs ainsi obtenues avec les équations qui donnent les valeurs de  $y(x + \frac{1}{2}\Pi)$ ,  $y(x + \frac{1}{2}T)$ , on trouve

$$L = 0, \quad M = \frac{\pi}{\text{mod.}(\nu\nu' - \nu'\nu)},$$

et enfin,

$$\frac{y(x + s\psi)}{y(x)} = e^{-\frac{\pi}{\text{mod.}(\nu\nu' - \nu'\nu)}} \Pi\left(1 + \frac{x}{(m,n) + s\psi}\right),$$

$$\text{mod.}(m, n) < T, \quad T = \infty.$$

Cela donne pour l'équation qu'il s'agissait d'établir; il ne faut pas remarquer qu'il y ait remarque que pour  $y = 0$ , on doit considérer à part le facteur  $1 + \frac{x}{(m,n)}$ , lequel multiplié par  $y(x)$  devient tout simplement  $x$ ; l'équation réduite donc dans ce cas.